NZ ROAD CRASH ANALYSIS

Power BI Portfolio Case Study

Waka Kotahi Crash Analysis System (CAS) | 2000–2025

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1. Business Problem

New Zealand has one of the higher road fatality rates among developed nations, placing significant human and economic cost on communities across the country. The Waka Kotahi Crash Analysis System (CAS) records every reported crash in New Zealand, providing a rich dataset spanning more than two decades.

This project addresses the following core question:

> *Where, when, and under what conditions are the most severe crashes occurring across New Zealand and what patterns should inform road safety investment?*

The resulting dashboard is designed to serve both executive stakeholders seeking high-level trends and operational analysts requiring granular regional and environmental detail.

2. Data Source

Dataset Overview	
Source	Waka Kotahi New Zealand Transport Agency: Crash Analysis System (CAS)
Access	Publicly available via the Waka Kotahi open data portal
Coverage	2000 to 2025 (25 years at time of project)
Volume	913,000 crash records
Original Columns	72 columns including location, severity, vehicle types, road conditions, and environmental factors
Coordinate System	NZTM2000 (EPSG:2193) New Zealand Transverse Mercator projection

3. Data Preparation

All data preparation was performed in Power Query within Power BI Desktop. The raw dataset required significant cleaning and transformation before it was suitable for analysis.

3.1 Column Reduction

The original dataset contained 72 columns, many of which were binary obstacle flags (e.g. bridge, cliffBank, debris, fence, guardRail) that were too granular to be analytically meaningful at a dashboard level. These were removed, reducing the working dataset to approximately 20 columns covering severity, location, road conditions, vehicle types, and environmental factors.

3.2 Data Type Standardisation

Column data types were explicitly set to ensure correct aggregation and filtering behaviour. Numeric fields such as crashYear, fatalCount, seriousInjuryCount, speedLimit, and vehicle type indicators were cast as integers. Text fields including crashSeverity, region, weatherA, roadSurface, and light were set as text type.

3.3 Null Value Handling

Null values were identified and handled in key categorical columns:

- crashSeverity nulls replaced with "Unknown"
- weatherA nulls replaced with "Unknown"
- region nulls replaced with "Unknown"
- speedLimit values of 0 were treated as null to prevent distortion of speed-based analysis

3.4 Coordinate Conversion (NZTM2000 → WGS84)

A significant technical challenge encountered during preparation was that the CAS dataset stores crash coordinates in NZTM2000 projection (EPSG:2193), using New Zealand Transverse Mercator easting and northing values. Power BI's map visuals require standard WGS84 latitude and longitude (EPSG:4326).

Sample raw coordinate values: X = 1,772,615 | Y = 5,896,529 (Auckland region). Direct use of these values placed all crash markers in the South Atlantic Ocean near South America.

An approximate conversion formula was applied via custom columns in Power Query:

```
Latitude = -36.8 + (([Y] - 5,896,529) / 111,320) - 0.3 Longitude =
174.7 + (([X] - 1,772,615) / 80,000) + 0.3
```

A bounding box filter was applied to exclude any coordinates falling outside New Zealand's geographic bounds (Latitude: -48 to -34, Longitude: 166 to 178), removing a small number of records with erroneous coordinate data.

Note: In a production environment, a precise NZTM2000 → WGS84 transformation would be performed using the pyproj Python library (EPSG:2193 → EPSG:4326), achieving metre-level accuracy across the full length of New Zealand.

4. Data Model

A star schema was implemented to optimise query performance and enable clean DAX measure writing. The model consists of one central fact table surrounded by dimension tables.

Data Model Structure	
CrashFact	Central fact table. One row per crash. Contains all numeric measures and foreign keys to dimension tables.
VehicleTypes	Disconnected table used to enable dynamic vehicle type analysis via SWITCH DAX measure. No relationship to fact table : intentional by design.

5. DAX Measures

All measures were organised in a dedicated Measures table following Power BI best practice. Key measures developed:

Core Measures	
Total Crashes	COUNTROWS(CrashFact) : count of all crash records
Total Fatalities	SUM(CrashFact[fatalCount]) : sum of fatalities across all crashes
Total Serious Injuries	SUM(CrashFact[seriousInjuryCount])
Crash Severity Rate	DIVIDE([Total Fatalities] + [Total Serious Injuries], [Total Crashes]) : proportion of crashes resulting in death or serious injury
YoY Crash Change	Year-on-year percentage change using SAMEPERIODLASTYEAR time intelligence function
Vehicle Crashes	SWITCH(SELECTEDVALUE(VehicleTypes[VehicleType]), ...) : dynamically returns crash count for the selected vehicle type using a disconnected table pattern

6. Dashboard Overview

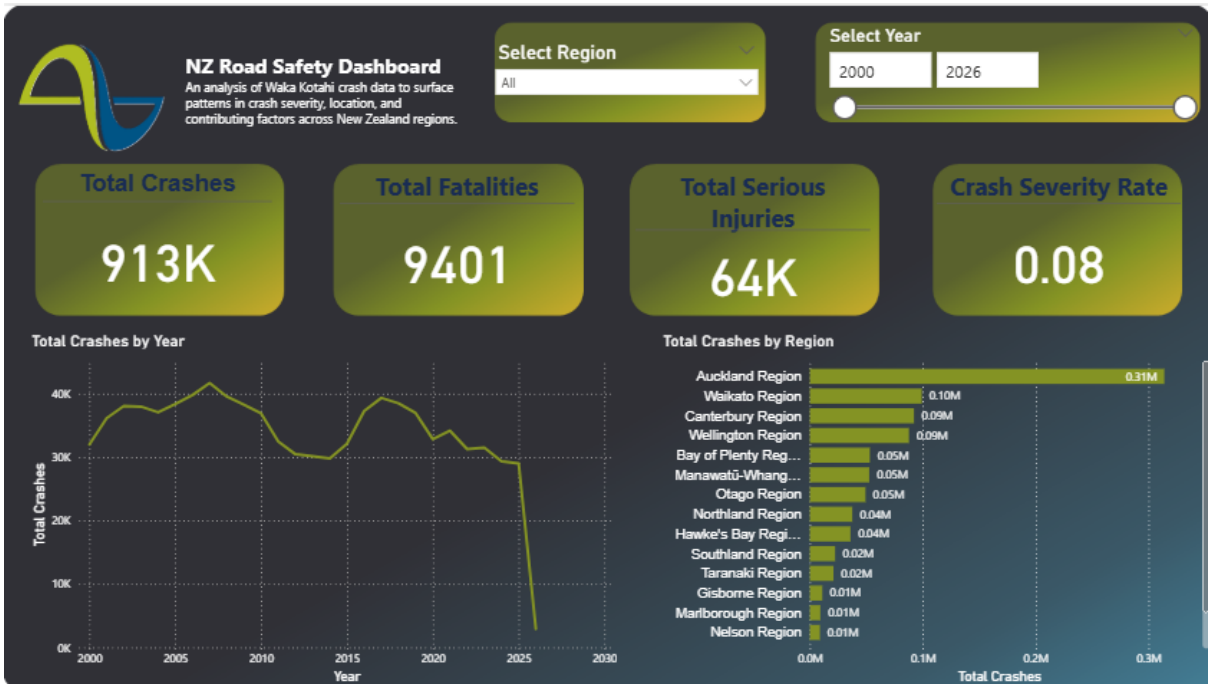
The dashboard comprises four report pages, designed to serve different analytical needs. Slicers are synchronised across pages where appropriate to maintain consistent filtering context.

Page 1: Executive Summary

Provides a high-level overview of NZ road crash trends for executive stakeholders.

Slicers: Region, Year.

- Four KPI cards: Total Crashes (913K), Total Fatalities (9,401), Total Serious Injuries (64K), Crash Severity Rate (8%)
- Line chart: Total crashes by year (2000–2025) showing long-term trend
- Bar chart: Total crashes by region, highlighting Auckland's disproportionate volume

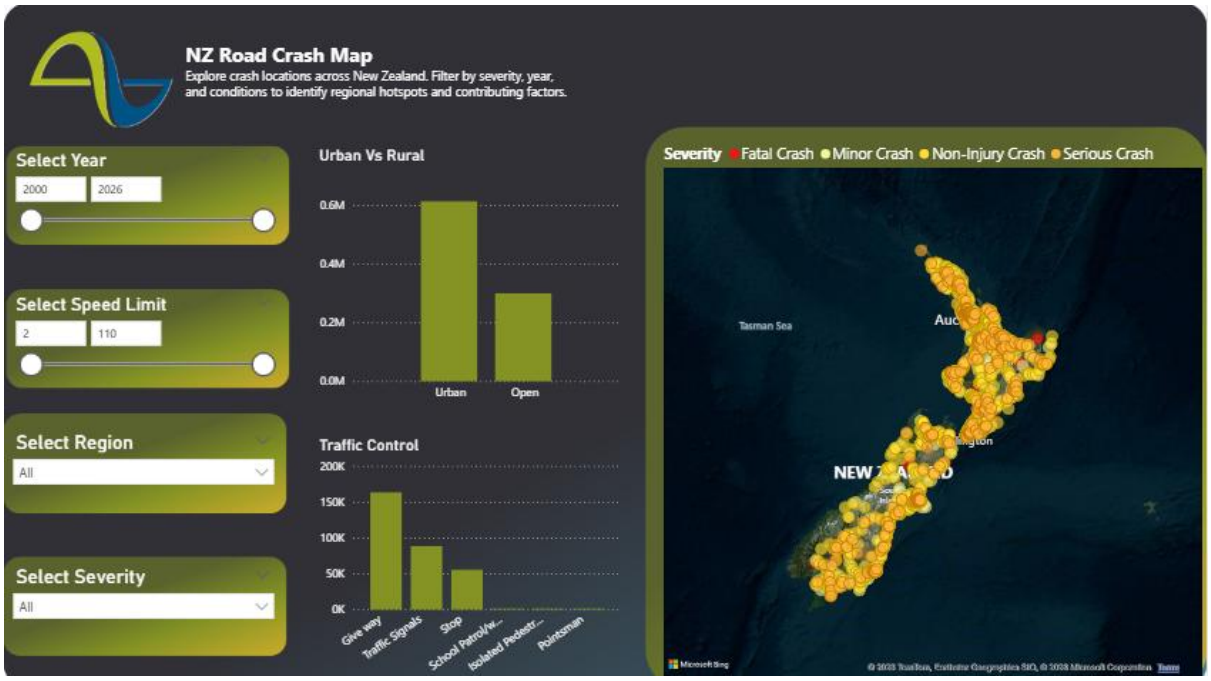


Page 2: Crash Map

Geographic hotspot analysis enabling spatial identification of high-risk areas.

Slicers: Year, Speed Limit, Region, Severity.

- Interactive map with crash location bubbles coloured by severity
- Urban vs open road bar chart
- Traffic control type bar chart



Page 3: Contributing Factors

Analysis of environmental and road conditions associated with crashes. Slicers: Region, Severity.

- Speed limit distribution chart
- Crash severity by road surface
- Treemap of crash distribution
- Weather conditions donut chart

Page 4: Road Environment

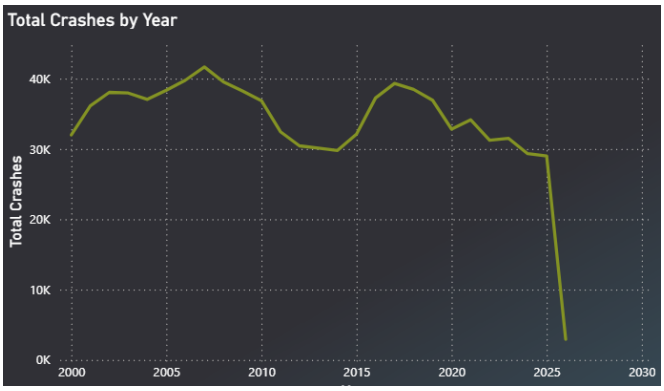
Analysis of road environment, lighting, terrain, and vehicle type involvement. Slicers: Severity, Region, Year.

- Light conditions bar chart (daylight vs dark vs dawn/dusk)
- Terrain chart (flat vs hill)
- Vehicle type involvement bar chart using disconnected table pattern
- Holiday period vs standard period crash comparison

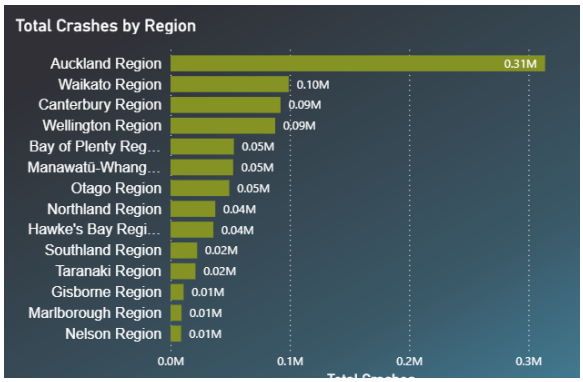
7. Key Findings

The following insights were derived from analysis of 913,000 crash records spanning 2000 to 2025:

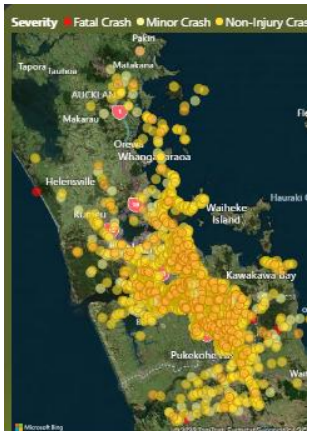
913K	9,401	64K	8%
Total Crashes	Fatalities	Serious Injuries	Severity Rate



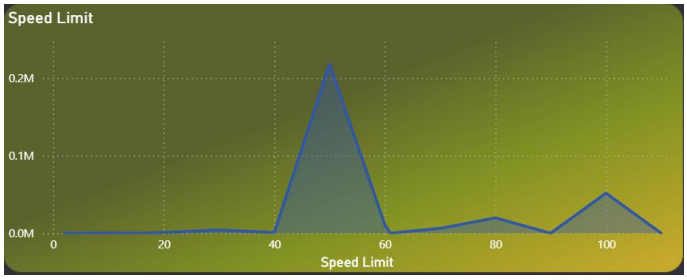
Crash volumes peaked in 2007 at 41,661 crashes, declined through 2011–2014 to the low 30,000s, rose again to a secondary peak of 39,314 in 2017, and have since trended downward to approximately 29,000 in 2025, suggesting road safety interventions over the past decade have had measurable impact.



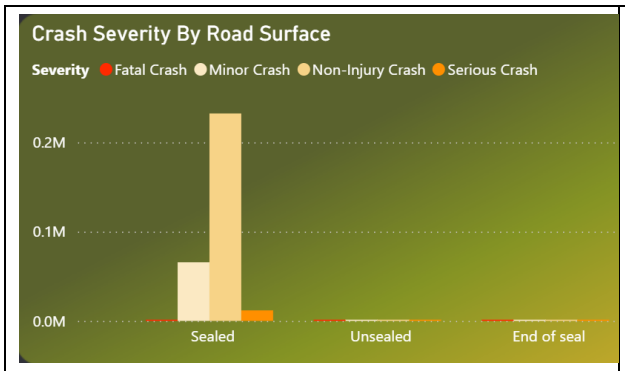
Auckland accounts for 0.31M crashes (34% of all crashes nationally), followed by Waikato (0.10M) and Canterbury (0.09M), reflecting both population density and traffic volume on state highways.

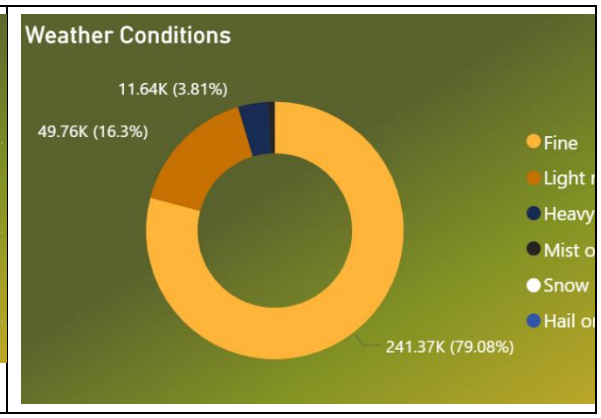


Crash hotspots are concentrated around Auckland's urban centre and major state highways, indicating that arterial road infrastructure and high-volume intersections are the primary risk environment.

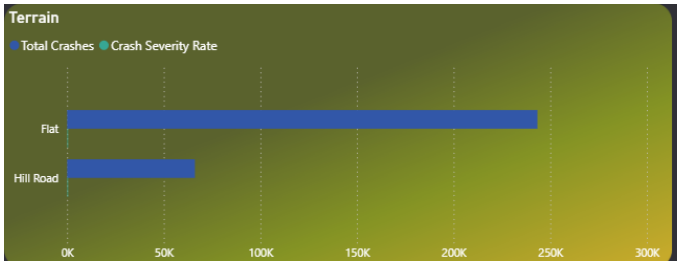


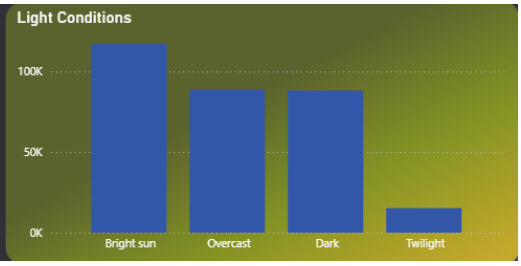
50 km/h speed limit zones account for the highest crash volume (500,000+), consistent with urban driving environments. However, 100 km/h zones (250,000+ crashes) carry disproportionate fatality risk given higher impact speeds.



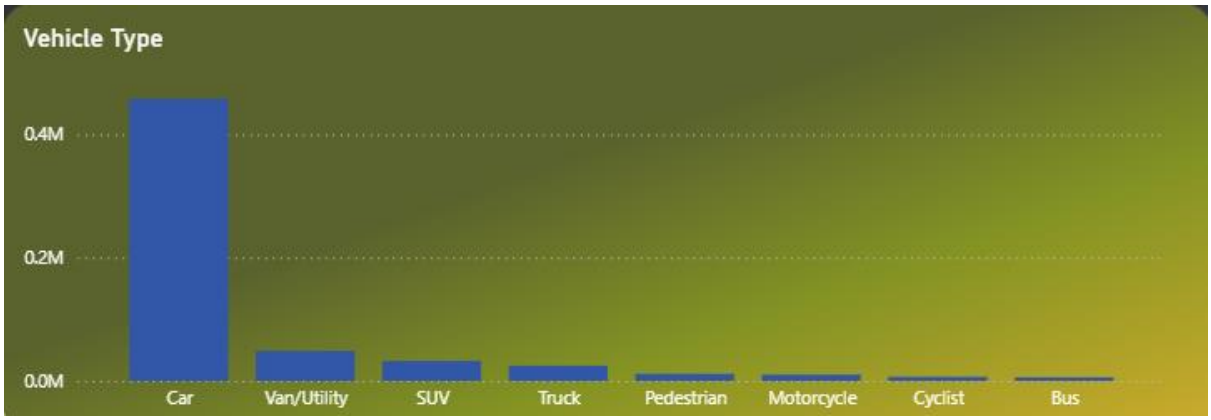


Fine weather and sealed road surfaces are the dominant conditions in crash records, indicating that driver behaviour rather than adverse conditions is the primary crash contributor in the majority of cases.

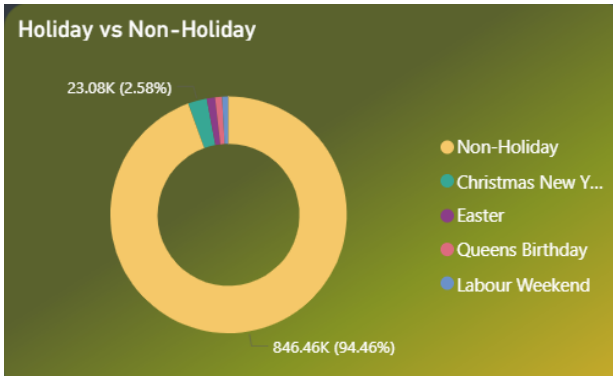




The majority of crashes occur in bright/sunny or overcast daylight conditions, and on flat terrain, reinforcing that crash risk is not confined to challenging environmental conditions.



Cars are by far the most commonly involved vehicle type at 450,000+ crashes, followed by vans/utilities (48,000) and SUVs (32,000). Vulnerable road users: pedestrians, cyclists, and motorcyclists, represent a smaller proportion of total volume but carry significantly higher severity rates given their lack of physical protection in a collision



Non-holiday periods account for the majority of crash volume; however, the Christmas/New Year holiday period shows elevated crash rates consistent with increased travel demand and driver fatigue.

8. Limitations

- Coordinate conversion from NZTM2000 to WGS84 was performed using an approximate formula. A production implementation would use the pyproj library for precise geodetic transformation across New Zealand's full geographic extent.
- The dataset records crash year only, no time of day or day of week data is available. A full datetime field would enable time-of-day heatmap analysis and day-of-week patterns, which are standard in road safety analysis.
- Crash records rely on Police reporting. Minor crashes not reported to Police are excluded from the dataset, meaning actual crash volumes are likely higher than recorded figures.
- Vehicle involvement columns (bicycle, pedestrian, motorcycle etc.) are binary flags indicating presence, not count. Multi-vehicle crashes involving multiple types are counted once per type, which may overstate involvement rates for individual categories.

9. Skills Demonstrated

Technical Skills	
Data Cleaning	Power Query: column reduction, null handling, data type standardisation, custom column creation
DAX	Calculated measures, time intelligence (SAMEPERIODLASTYEAR), SWITCH with disconnected tables, DIVIDE for rate calculations
Geospatial	NZTM2000 → WGS84 coordinate conversion, map visual with severity overlay, bounding box data quality filtering
Visualisation	Multi-page interactive report, synchronised slicers, KPI cards, bar/line/donut/treemap/map visuals
NZ Context	Waka Kotahi CAS dataset, NZ regional geography, NZTM2000 projection knowledge

Ngā mihi: Grace McNabb | 16 March 2026